AI-Driven Waste Classification: Comparative Analysis of Deep Learning Models (Custom CNN & ResNet-50) on Multiple Image Datasets for Smart Waste Management

***Abstract*—Reliable, scalable, and accurate automated classification systems are essential to addressing the global challenge of effective waste management. This paper offers a thorough comparative analysis of two deep learning techniques: a transfer learning-based ResNet-50 and a custom Convolutional Neural Network (CNN)—across four publicly accessible waste image datasets: TrashNet, Real-Waste (UCI), Boarkee, and the Kaggle Waste Classification dataset. In terms of sample size, visual complexity, and class balance, every dataset poses different difficulties. To guarantee strong generalization, both models were trained and assessed separately on every dataset utilizing stratified splits, comprehensive data augmentation, and early stopping.**

According to experimental results, ResNet-50 performs significantly better than the custom CNN in every scenario. ResNet-50's test accuracy was 92.89% versus 63.83% for the custom CNN on TrashNet, 91.59% versus 62.36% on RealWaste, 97.37% versus 67.11% on Boarkee, and 94.80% versus 87.90% on the large-scale Kaggle (binary) dataset. While the custom CNN performs poorly on imbalanced and multi-class datasets, per-class analysis shows that ResNet-50 not only performs better overall but also on minority and visually ambiguous classes, with per-class accuracies frequently surpassing 90%. These results are in line with recent research that emphasizes the benefits of deep architectures and transfer learning for waste classification tasks.

***Keywords*—Waste Classification, Deep Learning, ResNet-50, Transfer Learning, Circular Economy, E-Waste, Dataset Bias, Sustainable Development.**

I. Introduction

One of the most important environmental and socioeconomic issues of the twenty-first century is the growing global waste problem. Rapid urbanization, population growth, and changing consumption patterns are expected to drive the generation of municipal solid waste (MSW), which is expected to increase from 2.1 billion tonnes in 2023 to 3.8 billion tonnes by 2050 [1], [2]. The complexity and volume of contemporary waste streams make traditional waste management systems, which mainly rely on manual sorting and simple automation, increasingly insufficient. The prevalence of hazardous e-waste and composite materials [2], [5], [6] exacerbates these labor-intensive methods, which lead to consistently low global recycling rates (estimated at 17%) and a continued reliance on landfilling.

In waste recognition tasks, deep learning architectures—specifically, Convolutional Neural Networks (CNNs) and transfer learning frameworks like ResNet-50 support scalable solutions such as IoT-enabled smart bins with real-time monitoring, automated robotic sorting lines, and predictive analytics-optimized dynamic waste collection routing [3], [6], [9], [10].

There are difficulties in implementing deep learning in waste management. Insufficient sample sizes, class imbalance, and limited diversity plague many public datasets, limiting the generalizability and robustness of models [17].

Two deep learning models, a baseline Custom CNN and a transfer learning-based ResNet-50, are systematically evaluated across four diverse, publicly accessible waste image datasets in order to fill these gaps: TrashNet [12], RealWaste (UCI) [13], Boarkee [14], and the Kaggle Waste Classification dataset [15]. A strong testbed for comparing model performance is provided by the distinct challenges that each dataset poses in terms of sample size, class balance, and visual complexity. To guarantee strong generalization, both models were trained and assessed separately on every dataset using stratified splits and early stopping.

Our tests show that ResNet-50, which uses deep feature extraction and transfer learning, consistently performs better than the custom CNN on all datasets. Per-class analysis reveals that ResNet-50 performs better on minority and visually ambiguous classes in addition to having excellent overall accuracy. This finding is supported by recent research that highlights the significance of architectural depth and transfer learning for waste classification tasks [1], [7], [16], [17]. These results are further corroborated by per-class accuracy plots, and confusion matrices, which together demonstrate how important transfer learning and model complexity are to obtaining accurate, practical waste classification.

In this work, we aim to close the gap between algorithmic innovation and real-world implementation of AI-driven waste management systems by setting new or competitive baselines on underreported datasets and offering actionable insights into the interaction between model architecture, dataset complexity, and class distribution. Our findings complement the existing state of the art and provide a roadmap for further study and practical application to support the objectives of the circular economy and global sustainability [3], [6], [9].

II. Literature Review

The use of deep learning for automated waste classification has advanced quickly in recent years, with a strong emphasis on enhancing accuracy, efficiency, and deployability in the real world. The results of numerous recent studies are summarized in this review of the literature, which is arranged according to methodological trends and enduring difficulties.

A. Deep Learning Methods for Waste Categorization

Waste classification has greatly improved thanks to deep learning, which routinely outperforms conventional machine learning techniques. On a 16-category dataset, Qiao [1] showed that transformer-based models outperformed CNN, AlexNet, and ResNet variants, attaining a mean accuracy of 92.7%. The significance of model depth and pretraining was highlighted by Alalibo and Nwazor [5], who demonstrated that transfer learning with ResNet-50 on TrashNet produced a 92.06% accuracy in just seven epochs. A YOLOv8-based system for real-time, multi-class waste categorization was presented by Shrestha et al. [26]. They reported high precision and recall that made it appropriate for automated sorting. ECCDN-Net, as proposed by Islam et al. [27], increased organic waste classification efficiency and accuracy. WasteSegNet, a custom CNN trained on over 20,000 images, was developed by Saha et al. [28]. It outperformed conventional methods with an 87.5% test accuracy. Deep learning was applied to UAV imagery for large-area waste pile recognition by Liu et al. [29], who showed strong performance in a variety of environments. Kumar et al. [30] examined developments in CNNs, RNNs, and hybrid models, highlighting the advantages of big datasets and transfer learning. The efficiency of deep architectures across a range of waste types was confirmed by Islam et al. [31].

B. Architectural Innovations and Optimization

Using depthwise separable convolutions and redundancy-weighted feature fusion, Li et al. [7] enhanced ResNet-50, reducing model complexity and attaining 94.13% accuracy on TrashNet. Faster R-CNN models with deeper backbones (ResNet-101) performed better than shallower ones, particularly for small object detection, according to Ma'rifah et al. [23]. Islam et al. [27] combined parallel CNN branches and attention mechanisms in ECCDN-Net, enhancing interpretability and accuracy. Liu et al. [29] validated the practicality of deep networks for large-scale waste recognition. Saha et al. [28] demonstrated that custom architectures can outperform conventional methods, while Islam et al. [31] emphasized the significance of model selection and optimization.

C. Smart Waste Management and IoT Integration

González et al. [9] and Singh et al. [3] discussed smart bins with sensors and AI classifiers that track fill levels, recognize waste types, and improve collection efficiency while reducing operational costs and environmental impact. Tele2IoT [32] and RecyclingInside [10] emphasized the use of IoT for predictive maintenance, dynamic scheduling, and real-time data collection. Using mobile apps and smart sorting stations, Bridgera [33] demonstrated IoT-enabled analytics and public engagement. AT&T Business [35] talked about how combining IoT and AI can improve sustainability and lessen reliance on hardware. IoT-enhanced deep learning models in urban waste management have been shown to improve efficiency and public health (Islam et al. [27] and Saha et al. [28]).

D. Primary Difficulties and Unresolved Problems

Table I provides a summary of the main obstacles and unresolved problems in deep learning-based waste classification.

TABLE I: Key Challenges and Open Issues in Deep Learning-Based Waste Classification

Challenge	**Description**
Dataset Imbalance and Diversity	Many public datasets are limited in size and diversity, leading to class imbalance and reduced model robustness [1], [5], [7], [17], [22], [27], [30].
Generalization	Models often struggle to generalize to novel waste streams, new materials, or changing environmental conditions [1], [22], [30], [31].
Computational Demands	Deep architectures, while accurate, can be resource-intensive, posing barriers for deployment on edge devices or in resource-constrained settings [7], [27], [30].
Limited Annotated Data	The scarcity of large-scale, well-annotated datasets remains a bottleneck for training robust deep learning models [27], [30], [31].
Model Interpretability	Deep learning models are often treated as black boxes, making it difficult to interpret decisions and build stakeholder trust [27], [30].
Integration with IoT	Challenges include data interoperability, connectivity, and real-time processing when deploying deep learning models within IoT frameworks [10], [32], [33], [35].

E. Summary

In summary, the literature demonstrates that deep learning—especially transfer learning, attention mechanisms, and hybrid models—has greatly advanced automated waste classification. Continued research and collaboration are crucial for realizing sustainable, scalable solutions in waste management.

III. Methodology

A. Pipeline Overview

This study uses a reproducible deep learning pipeline for waste image classification. As summed up in Table II, the workflow is made for reliable benchmarking, equitable comparison.

TABLE II: Pipeline Components and Tools

Step	Library/Tool	Purpose
Data Loading	PyTorch, PIL	Image I/O, folder structure
Data Augmentation	torchvision	Resize, flip, rotate, color jitter, normalize
Stratified Split	scikit-learn	Balanced train/val/test splits
Model Building	PyTorch, torchvision.models	Custom CNN, ResNet-50
Training/Early Stopping	PyTorch	Efficient, avoids overfitting
Evaluation/ Visualization	sklearn, matplotlib, seaborn	Metrics, confusion matrix, t-SNE
Device Management	PyTorch	GPU/CPU acceleration

B. Datasets

Four different datasets that each presented different difficulties in terms of sample size, visual complexity, and class balance were used to benchmark the models. The properties and splits of the dataset are summarized in Table III.

TABLE III: Overview of Waste Classification Datasets

Dataset	Classes	Images	Train/Val/Test	Source
TrashNet [12]	6	2,527	1,768/253/506	GitHub
RealWaste [13]	9	4,752	3,326/475/951	UCI
Boarkee [14]	5	377	263/38/76	GitHub
Kaggle Waste [15]	2	25,077	17,553/2,508/5,016	Kaggle

Class Distributions: Distribution of Classes: The total number of images available for each class in the dataset is indicated by the number in parenthesis.

- **TrashNet**: Cardboard (403), Glass (501), Metal (410), Paper (594), Plastic (482), Trash (137)
- **RealWaste**: Cardboard (461), Food Organics (411), Glass (420), Metal (790), Misc. Trash (495), Paper (500), Plastic (921), Textile Trash (318), Vegetation (436)
- **Boarkee**: Glass (70), Metal (73), Organic (76), Paper (70), Plastic (88)
- **Kaggle Waste**: Organic (13,966), Recyclable (11,111)

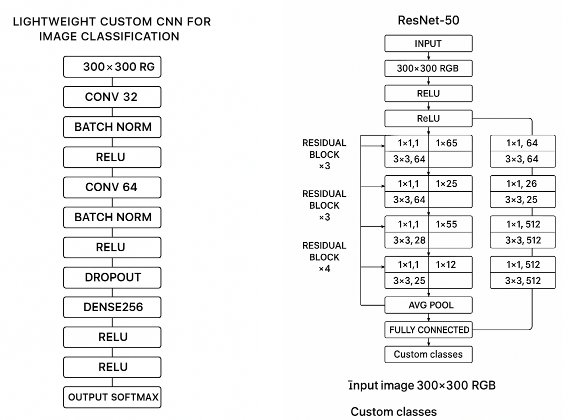
C. Data Handling

Every picture was resized to 300 by 300 pixels. Stratified sampling for every dataset guaranteed balanced train/val/test splits (usually 70%/10%/20%), which is essential for an equitable assessment with unequal classes.

D. Model Architectures

A lightweight custom Convolutional Neural Network (CNN) and a transfer learning-based ResNet-50 are the two main deep learning models for waste image classification that are evaluated in this study 1a.

Adaptive learning rates and gradual unfreezing are used to fine-tune the ResNet-50 model, which is a 50-layer deep residual network that was pretrained on ImageNet. The last layer is changed for every dataset 1b.



(a) Custom CNN Architecture (b) ResNet-50 Architecture

Fig. 1: Architectural diagrams of the two models used in this study.

E. Training Protocol

To guarantee computational efficiency, model training and inference were carried out on GPUs (Tesla T4, RTX 3050). For better generalization, weight decay was applied specifically to ResNet-50 using the Adam optimizer. Up to 50 epochs of training were conducted, and if validation loss did not improve for five consecutive epochs, early stopping was initiated. For the final assessment, the top-performing epoch was chosen (see Table IV).

F. Epochs and Early Stopping

A maximum of 50 epochs per model were used for training; however, early stopping usually decreased this (see Table IV).

TABLE IV: Best Epochs Selected by Early Stopping

Dataset	Custom CNN	ResNet-50
TrashNet	14	9
RealWaste	29	9
Boarkee	11	14
Kaggle Waste	21	2

G. Visualization and Reporting

Overfitting was detected with the aid of accuracy and loss curves, which monitored training progress. Per-class accuracy plots and confusion matrices offered comprehensive information on the model's advantages and disadvantages for each class. Furthermore, model performance was compared across datasets using bar charts, which together provided strong benchmarking and lucid results presentation.

H. Summary of Achievements

- **Reproducible pipeline** for multi-dataset waste classification
- **Consistent stratified splitting and evaluation**
- **Early stopping and checkpointing** for efficient training
- **Comprehensive metrics and visualizations** for robust comparison
- **Scalable codebase** supporting additional datasets and models

By creating new empirical baselines and offering useful insights for further study and implementation, this methodology guarantees an impartial, open, and comprehensive evaluation of deep learning approaches in waste classification.

IV. Results and Discussion

This section presents a thorough analysis of the Custom CNN and ResNet-50 models on four benchmark waste classification datasets: Kaggle Waste [15], RealWaste [13], Boarkee [14], and TrashNet [12]. We present learning curves, confusion matrices, and overall and per-class accuracy for each dataset.

The Custom CNN's test accuracy on TrashNet was 63.83%, while its per-class accuracy ranged from 40.21% (plastic) to 91.60% (paper). ResNet-50, on the other hand, used transfer learning to achieve a significantly higher test accuracy of 92.89%, with all per-class accuracies surpassing 77.78% V.

The baseline CNN only achieved 62.36% test accuracy on the RealWaste dataset, which had nine classes and a noticeable class imbalance. As demonstrated in figures 2, ResNet-50 once again outperformed the baseline, attaining 91.59% test accuracy and per-class accuracies above 82% for all major categories.

Both models performed better on the small and balanced Boarkee dataset, but the difference was still significant: the Custom CNN's accuracy was 67.11%, while ResNet-50's was 97.37%.

Both models showed good performance on the large binary Kaggle Waste dataset, but ResNet-50 continued to have a distinct advantage: 94.80% test accuracy compared to 87.90% for the Custom CNN 3.

As shown in Table V, the cross-dataset model comparison shows that ResNet-50 always does better than the custom CNN on all four benchmark datasets.

Figure 4 shows a direct, side-by-side comparison of how well each model works on each dataset. This makes it easy to see the difference in performance between the architectures.

TABLE V: Test Accuracy and Per-Class Performance Across Datasets

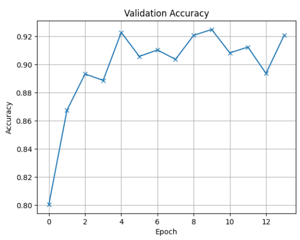
Dataset	Model	Accuracy (%)	Per-Class Accuracy (Low / High)
TrashNet	Custom CNN	63.83	40.21–91.60
	ResNet-50	92.89	77.78–98.78
RealWaste	Custom CNN	62.36	34.38–91.95
	ResNet-50	91.59	82.61–98.85
Boarkee	Custom CNN	67.11	42.86–86.67
	ResNet-50	97.37	92.86–100.00
Kaggle	Custom CNN	87.90	86.28–89.76
	ResNet-50	94.80	90.78–97.60

A. Key Observations

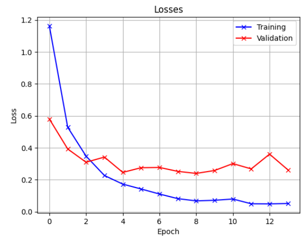
- **Transfer learning with ResNet-50 consistently outperformed the custom CNN** by large margins (up to 30% absolute accuracy) on complex, multi-class datasets (TrashNet, RealWaste).
- **Custom CNN performance was competitive only on the binary Kaggle dataset**, but still trailed ResNet-50 by nearly 7%.
- **Per-class analysis and confusion matrices** revealed that minority and visually ambiguous classes (e.g., trash, textile, glass) were most challenging for the baseline CNN, while ResNet-50 maintained robust performance.
- **Early stopping and validation curves** confirmed that deeper models converged faster and were less prone to overfitting, especially when using adaptive learning rates and gradual unfreezing.

B. Discussion

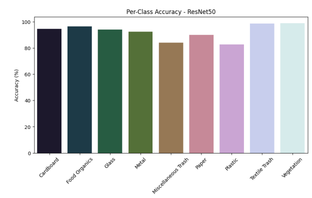
The findings unequivocally show that deep transfer learning architectures, like ResNet-50, are highly advised for classifying waste in the real world, especially when there is intra-class variation and class imbalance. Despite its computational efficiency, the custom CNN had limited generalization, particularly when dealing with



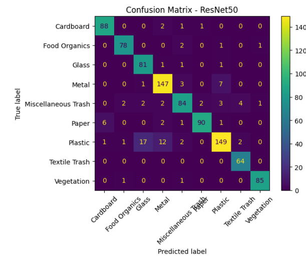
(a) Accuracy vs. Epochs



(b) Loss vs. Epochs

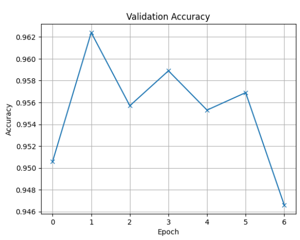


(c) Per-Class Accuracy

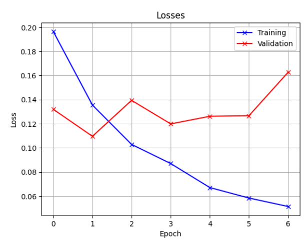


(d) Confusion Matrix

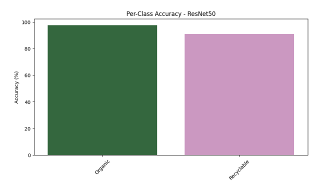
Fig. 2: ResNet-50 results on RealWaste: (a) Accuracy vs. Epochs, (b) Loss vs. Epochs, (c) Per-Class Accuracy, (d) Confusion Matrix



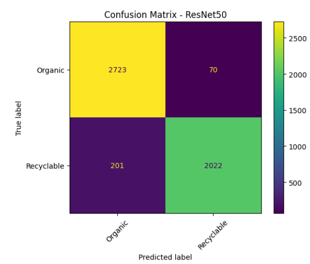
(a) Accuracy vs. Epochs



(b) Loss vs. Epochs

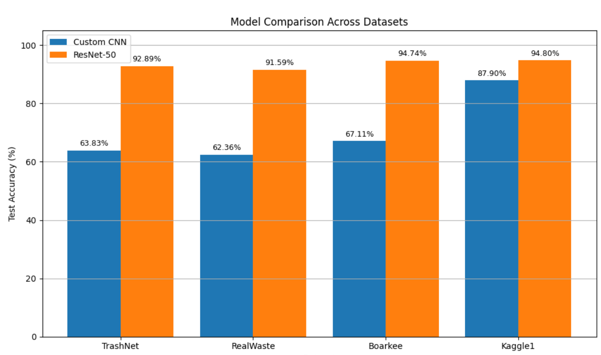


(c) Per-Class Accuracy

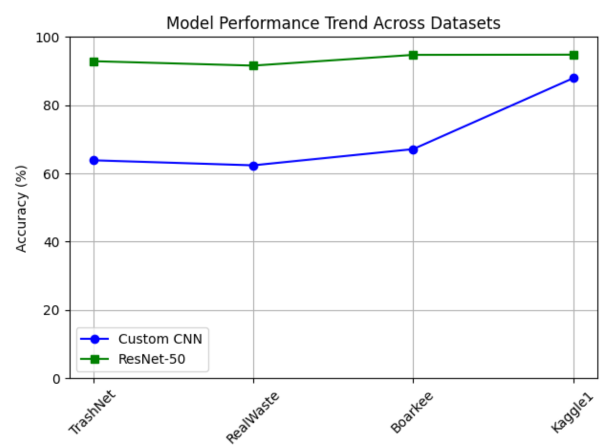


(d) Confusion Matrix

Fig. 3: ResNet-50 results on Kaggle Waste: (a) Accuracy vs. Epochs, (b) Loss vs. Epochs, (c) Per-Class Accuracy, (d) Confusion Matrix



(a) Overall accuracy comparison of Custom CNN and ResNet-50 across all datasets.



(b) Overall model performance of Custom CNN and ResNet-50 across all datasets.

Fig. 4: Comparative evaluation of Custom CNN and ResNet-50 across multiple datasets.

imbalanced and multi-class datasets. These results are in line with recent research [1], [5], [7], which emphasizes the significance of dataset diversity, transfer learning, and model depth for reliable waste classification.

By combining accuracy/loss curves, per-class accuracy and confusion matrices, the thorough visualization approach used allows for clear benchmarking and useful understanding of the model's advantages and disadvantages.

In summary, ResNet-50, which uses deep feature extraction and transfer learning, is the obvious option for reliable and accurate waste classification in a variety of situations. For binary or small-scale tasks, custom CNNs might be adequate, but they need to be improved further to manage class imbalance and visual ambiguity.

V. Conclusion

Using four different datasets, this study shows that transfer learning with ResNet-50 offers a definite advantage over a custom CNN for waste classification, with higher accuracy and robustness. ResNet-50 exhibits strong per-class accuracy and dependable generalization, even for visually similar and minority waste types, and consistently outperforms the custom CNN, particularly in complex and unbalanced scenarios. In contrast, the custom CNN struggles with class imbalance and category overlap but performs well on simpler or binary tasks. These results highlight the significance of pretraining and deep architectures for practical, efficient waste management solutions. Addressing class imbalance, increasing the diversity of datasets, and refining models for edge deployment. Scalable, precise, and useful AI-driven waste management solutions will be made possible by developments in data augmentation, lightweight transfer learning, and multi-modal sensing.

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